Precision Tissue Mapping from Spatial Transcriptomics Using Graph Learning

A neighborhood-aware platform for discovering spatially localized molecular pathology

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Research Proposal for Mentor Review | June 15, 2026

The problem. Spatial transcriptomics measures gene expression while keeping track of where each measurement sits in the tissue (a spot or a single cell, depending on the platform). This lets us ask not only which genes change in disease, but where in the tissue the change occurs, a central question in precision oncology and neuropathology [1]. Before a spatial slide can be interpreted, however, every measured spot must first be assigned to its tissue domain (a cortical layer, a tumor region, an immune niche). Today that assignment relies on slow, subjective manual annotation, or on expression-only clustering that ignores location and produces noisy, spatially incoherent maps. This project builds a neighbourhood-aware tissue-mapping pipeline that turns a spatial slide into an interpretable molecular map, then uses that map to localize where disease-relevant biology appears, while testing whether the spatial neighbourhood is what makes the map work.

Preliminary results (pilot on two public datasets)

Initial results on two public datasets are encouraging. On the LIBD DLPFC benchmark (12 Visium sections, 47,329 expert-annotated spots, 7 cortical domains), our spatial GNN reaches macro-F1 0.75 and ARI 0.65 under the harder cross-section split (whole tissues held out, 3 seeds), against 0.52 for XGBoost and 0.49 for an MLP trained on the same expression features, a gain of +0.24 macro-F1. The result holds across architectures: a GraphSAGE encoder and a graph-attention (GAT) encoder both reach about 0.75 F1. I also tested whether this was specific to Visium. On single-cell osmFISH mouse cortex (33 genes), the GNN reaches macro-F1 0.96 against 0.54 (XGBoost) and 0.63 (MLP). A graph-removal test on both tissues collapses the same GNN back to the blind MLP, which indicates the gain comes from spatial structure rather than from added model capacity. The proposal below formalizes, stress-tests, and extends these results toward tumor-microenvironment mapping.

Why this project. I chose tissue mapping because it sits where neuroscience, cancer biology, and machine learning meet, and because the question is concrete: can a model tell which part of a tissue a measurement came from? What surprised me in the pilot was how much the answer depends on context. Letting the model look at a spot's neighbours, rather than the spot on its own, raised macro-F1 from about 0.5 to 0.75 on held-out brain sections. Understanding and stress-testing that gap is what the project is about.

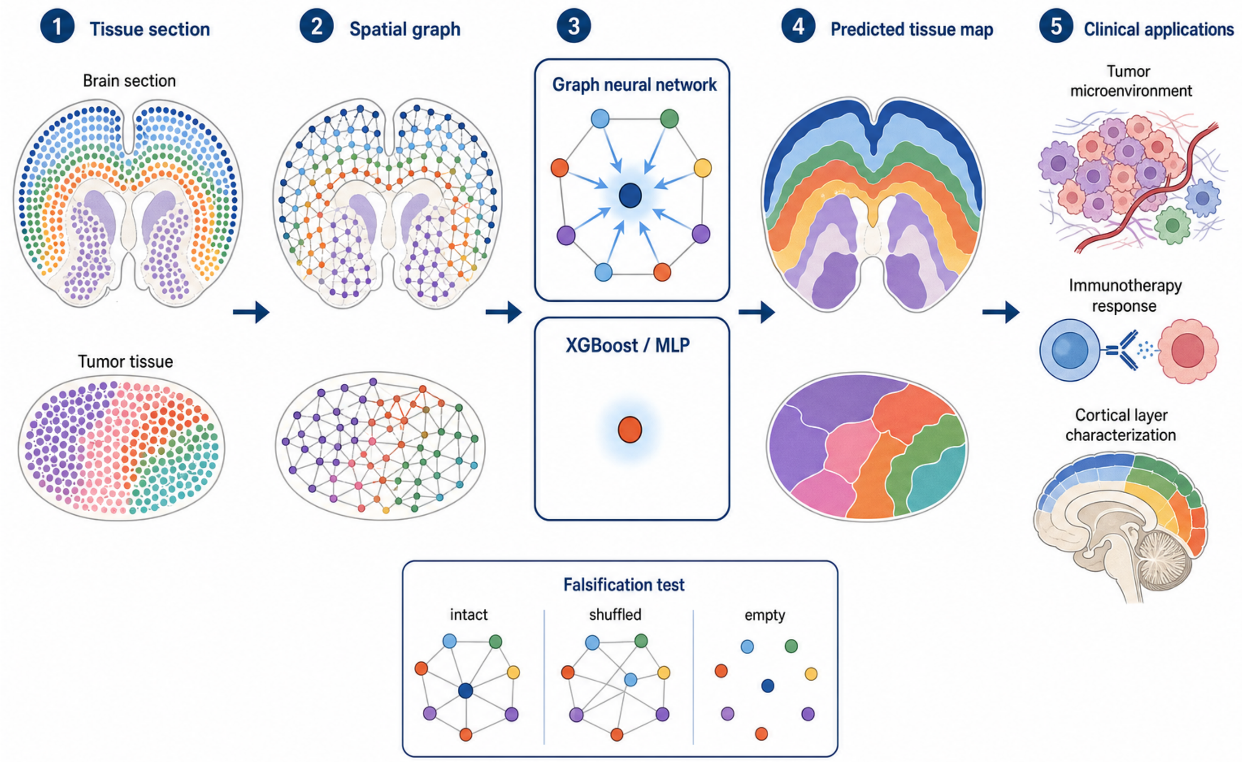

Figure 1: Study overview. Each spatial-transcriptomics section (human cortex or tumor tissue) is turned into a graph whose nodes are spots or cells and whose edges connect spatial neighbours. A graph neural network labels each spot by pooling its neighbourhood, while XGBoost and MLP baselines see the same expression features but treat each spot in isolation. We compare the predicted tissue maps, run a falsification test that shuffles or removes the graph, and connect the method to downstream uses in oncology and neuropathology.

1 Background and Significance

Why spatial context is clinically important. Tissue function depends on how cells are arranged in space. In the cerebral cortex, six layers (L1–L6) above the white matter each carry distinct cell types and gene programs. In a tumor, malignant cells, stroma, blood vessels, and infiltrating immune cells form spatially structured niches. Disease changes are often spatially localized: tau pathology in Alzheimer's follows a laminar gradient, schizophrenia shows layer-specific cortical deficits, and a tumor's response to immunotherapy depends on where immune cells sit relative to the malignant front. Standard bulk RNA sequencing averages this away. Spatial transcriptomics (10x Visium, MERFISH, Xenium, CosMx, Stereo-seq) preserves it, and a 2024 review in the Journal of Cancer Research and Clinical Oncology argues that this spatial information can advance precision oncology by capturing tumor heterogeneity, metastasis, prognosis, and immunotherapy response that bulk profiling misses [1].

Why it matters	Consequence
Tissue function is spatially organized	Disease changes are region/layer-specific, not diffuse.
Spatial omics preserves location	We can finally ask *where* pathology starts.
But spots must be assigned to a domain first	Domain mapping is an important early step for most downstream spatial analyses.

Table 1: Accurate spatial-domain mapping is the step that turns a spatial-transcriptomics slide into interpretable, region-resolved biology, in both brain disease and cancer.

2 The Computational Gap

Domain mapping is hard for two complementary reasons, and current practice addresses neither well.

Gap 1. Manual annotation does not scale. Expert annotation of a single section is slow and subjective. A disease cohort spans dozens of sections across many donors, where hand-labelling becomes the bottleneck.

Gap 2. Per-spot expression is weak and structure-blind. A single spot's expression is sparse and dropout-corrupted, so models that classify each spot in isolation (XGBoost, MLP) reach an accuracy ceiling and yield spatially fragmented domains.

The observation behind our approach is straightforward: a spot's domain is largely a property of its spatial neighbourhood. Adjacent spots usually share a domain, so the signal missing from one noisy spot is present in the tissue around it (Fig. 2). A model that can use the neighbourhood should therefore do better than one that cannot, and the size of that gap is a direct measure of how much the spatial structure is worth.

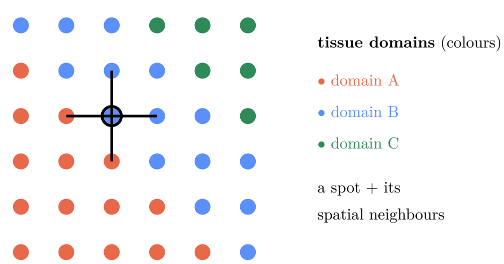

Figure 2: Domains are spatially contiguous, so a spot's label tends to agree with its neighbours'. A single noisy spot may be ambiguous, but its neighbourhood reveals its domain, which is the signal a spatial model can use and a per-spot model cannot.

3 Central Hypothesis and Specific Aims

Central hypothesis. A spot's tissue domain is encoded not only in its own gene expression but in its spatial neighbourhood. A model that uses that neighbourhood should therefore produce more accurate tissue maps than expression-only models, and removing the spatial graph should erase the gain.

Specific Aims

Aim 1 (Platform): a neighbourhood-aware tissue-mapping pipeline. Build a reproducible pipeline that converts a spatial-transcriptomics slide into an interpretable tissue-domain map, using each spot's gene expression together with its local spatial neighbourhood, and apply it across technologies (Visium, osmFISH, and a tumor section).

Aim 2 (Validation): a spatial-validity test. Check whether the maps depend on real tissue structure: compare against tuned non-graph baselines (XGBoost and MLP) on identical features, then shuffle or remove the neighbourhood graph as a falsification control.

Aim 3 (Discovery): a biological discovery layer. Recover known marker genes, delineate domain boundaries, and surface spots where the map confidently disagrees with manual annotations, which are candidate annotation errors or unrecognized micro-domains for expert review.

4 Relationship to Prior Work

GNN-based spatial-domain identification is an established field. SpaGCN [7], STAGATE [6], and GraphST [5] all apply graph neural networks to spatial-omics data and are standard baselines on the DLPFC benchmark. We therefore do not claim a new architecture. Those methods are predominantly unsupervised (graph autoencoders followed by clustering, scored by ARI against manual labels) and are compared mainly against other clustering methods. Our contribution is a complementary and, in this field, under-asked question: how much does spatial structure actually add over strong non-graph models given the same features? We answer it with (i) a supervised, leakage-safe cross-section protocol (train on whole sections, predict held-out ones, which is the disease-cohort regime); (ii) tuned XGBoost and MLP baselines on the same features, which spatial-domain papers rarely report; (iii) a graph-removal test that separates the spatial signal from model capacity; and (iv) replication across two technologies (Visium and osmFISH). The deliverable is a careful estimate of the value of tissue structure. It is a rigorous benchmark and analysis, not a state-of-the-art or novel-method claim.

5 Datasets

Dataset	Content	Role
LIBD DLPFC (10x Visium) [2, 3]	12 sections, 47,329 spots, expert manual layer labels (L1–L6 + WM)	Primary ground-truth benchmark (cross-section)
osmFISH mouse cortex [4]	single-cell resolution, 33 genes, ∼4,839 cells, region labels	Second technology / generalization
Controlled synthetic tissue	domains defined purely by neighbourhood; per-spot signal weak by construction	Mechanism isolation / sanity check
Tumor ST (Visium/Xenium, stretch)	public solid-tumor sections with niche annotations	Clinical extension (Aim 3)

Table 2: All data are public and de-identified. LIBD DLPFC is the field-standard ground truth for spatial-domain methods. osmFISH tests a different assay with far fewer genes, where per-cell features are weakest.

6 Methodology

From tissue to spatial graph. Each section becomes a graph: nodes are spots or cells, node features are preprocessed expression (normalize, log1p, highly-variable genes, then PCA), and each node is connected to its k nearest spatial neighbours. The 2D coordinates are used only to build the graph, never as model features, so any GNN benefit is attributable to message passing over tissue structure rather than to memorizing absolute position.

6.1 Model: a spatial graph neural network

The GNN classifies each spot by pooling its neighbourhood over L rounds of message passing. With a GraphSAGE encoder,

$$h_v^{(l)} = \sigma\Big(W^{(l)}\big[\, h_v^{(l-1)} \,\|\, \mathrm{AGG}_{u\in\mathcal{N}(v)}\, h_u^{(l-1)} \,\big]\Big), \qquad \hat{y}_v = \mathrm{softmax}\big(W_o\, h_v^{(L)}\big), \tag{1}$$

trained with cross-entropy on annotated spots. Aggregation denoises the sparse per-spot signal, which is the property Gap 2 lacks. We compare **GraphSAGE**, **GAT** (graph attention, which learns per-neighbour weights), and GCN encoders; all use PyTorch Geometric.

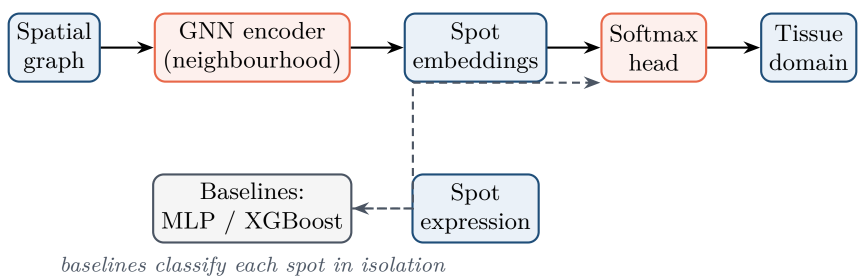

Figure 3: Pipeline. The GNN (orange) pools each spot's spatial neighbourhood; the non-spatial baselines (gray) classify each spot alone from the same features. The only difference between them is access to the spatial graph.

6.2 Baselines and leakage-safe evaluation

We benchmark against tuned XGBoost and an MLP on the same expression features, so the comparison isolates the value of tissue structure. We use three splits: (i) *cross-section*, where whole sections are held out for test (layer layouts differ per section, so absolute position cannot transfer and only neighbourhood-relative reasoning generalizes; this is the disease-cohort regime and our headline test); (ii) *within-section*, where contiguous spatial blocks are held out; and (iii) *stratified*, a standard transductive node split for single-section data (osmFISH). In the cross-section disjoint-union graph, train and test sections are separate connected components, so message passing cannot leak labels across the split. We report accuracy, macro-F1 (domains are imbalanced), and the adjusted Rand index (ARI, the field-standard domain metric), as mean ± std over seeds.

7 Preliminary Results

A working pilot already supports the claims above.

Macro-F1 / ARI	XGBoost	MLP	SAGE	GAT
DLPFC, *cross-section* (3 seeds)	0.52 / 0.37	0.49 / 0.38	**0.75 / 0.65**	**0.74 / 0.66**
osmFISH, *within-section* (3 seeds)	0.54 / 0.44	0.63 / 0.51	**0.96 / 0.95**	**0.97 / 0.95**

Table 3: Spatial GNNs beat the non-spatial baselines on the same features, on two tissues and two technologies. The osmFISH gap is larger because its 33-gene per-cell signal is weakest, so the neighbourhood matters most. osmFISH is a single section, so this is a transductive within-section split, an easier regime than DLPFC's cross-section test, and is reported as such.

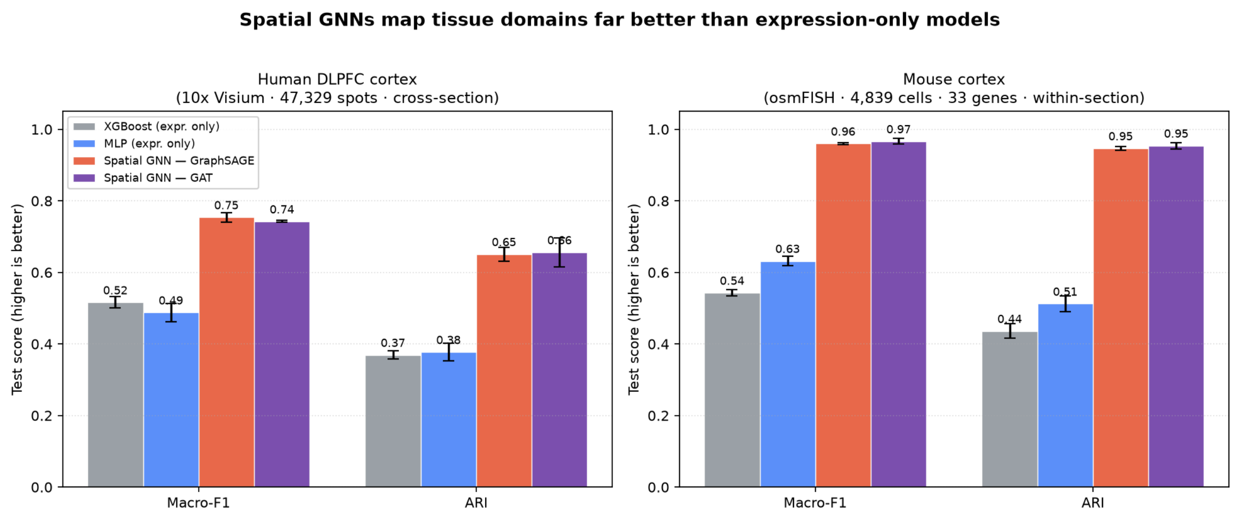

Figure 4: Headline comparison. On both human DLPFC (Visium) and mouse cortex (osmFISH), the spatial GNN (GraphSAGE and GAT) clearly outperforms XGBoost and MLP, which see the same expression features but not the tissue graph.

8 Rigor: Falsification and Biological Interpretation

8.1 Is the spatial structure really doing the work?

We run the same GNN on three graphs. If the gain is genuinely spatial, destroying the tissue structure should erase it, and it does, on both tissues (Fig. 5, Table 4).

GNN macro-F1	Intact graph	Shuffled edges	Empty graph	MLP ref.
DLPFC (cross-section)	**0.75**	0.49	0.49	0.48
osmFISH (within-section)	**0.95**	0.58	0.61	0.63

Table 4: Falsification test. Replacing the true spatial kNN graph with degree-matched random edges, or with no edges, collapses the GNN onto the blind MLP on both tissues. The advantage comes from the graph, not the network capacity.

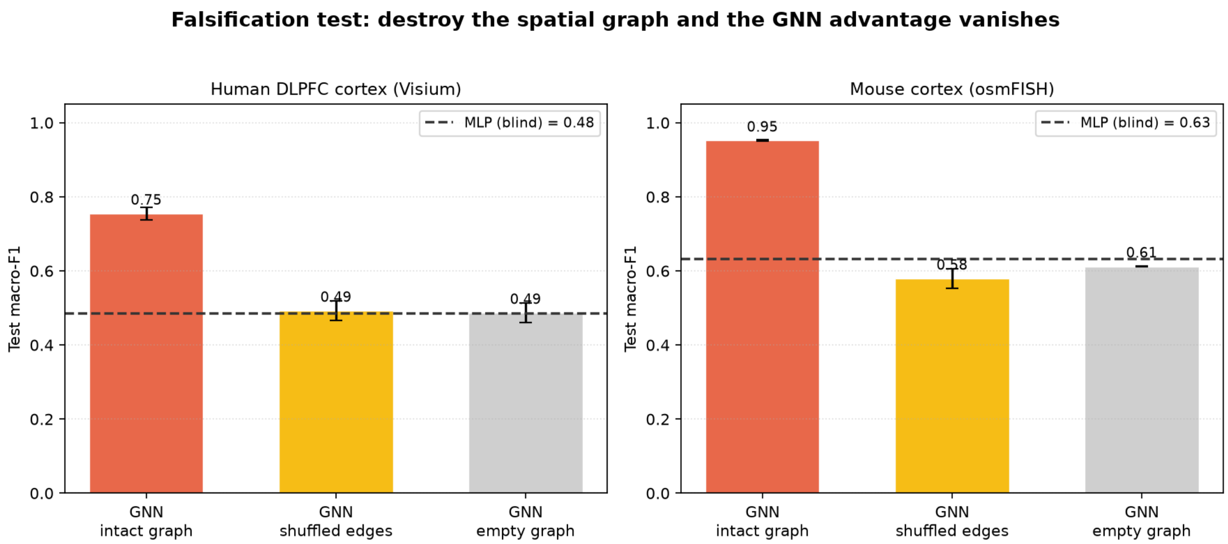

Figure 5: The same falsification test, visualized on both tissues. Only the intact spatial graph clears the MLP reference line; shuffled and empty graphs fall back to it.

8.2 From accuracy to biology

For each predicted domain we will (i) extract the genes driving predictions and confirm recovery of known marker genes (cortical-layer markers for DLPFC), and (ii) flag spots where the model confidently disagrees with the manual label, which are candidate annotation errors or genuine micro-domains for expert review. This turns an accuracy number into a hypothesis a biologist can act on.

9 Clinical Translation: Toward Precision Pathology

The cortical-layer benchmark serves as the proof of concept. The same capability, accurate and automated domain mapping that stays spatially coherent, is one requirement for translating spatial transcriptomics to the clinic, particularly in oncology [1].

Application	How domain mapping helps
Tumor microenvironment	Separate tumor core, invasive front, stroma, and immune niches to study heterogeneity and metastasis.
Immunotherapy response	Localize where immune cells sit relative to malignant cells, a spatial correlate of response.
Neuro-/neurodegenerative disease	Pinpoint which cortical layers carry early molecular pathology (Alzheimer's, schizophrenia).
Pathologist support	Annotate complex molecular slides faster and more reproducibly than manual labelling.

Table 5: The brain benchmark tests the method; precision oncology and pathology are where it pays off. All predictions are research-only and hypothesis-generating.

10 Project Strengths and Feasibility

A few things give me confidence that the project is both feasible and worthwhile (Table 6): it uses real, public data; it has a clear medical motivation; it compares against tuned non-graph baselines under leakage-safe splits; it includes a causal check on where the gain comes from; and it ties the accuracy back to known biology.

Criterion	Commitment
Real, sizable public data	LIBD DLPFC: 12 sections, 47k annotated spots, plus a second technology (osmFISH).
Clear medical motivation	Spatially-resolved localization of disease pathology; precision oncology bridge.
Rigorous evaluation	Tuned XGBoost and MLP baselines; leakage-safe splits; multi-seed macro-F1 and ARI.
Causal / falsification test	Graph-removal ablation on two tissues collapses the GNN to the MLP.
Interpretability → discovery	Recover marker genes; surface candidate annotation errors.
Transparent reporting	Any narrowing of the gap (e.g. easier transductive splits) is reported as-is.

Table 6: Strengths and feasibility of the project.

11 Eight-Week Timeline

The pilot already de-risks the core pipeline. The eight-week plan focuses on validation, ablations, biological interpretation, and the tumor-ST extension (Table 7).

Weeks	Milestones
1–2	Data pipeline (DLPFC + osmFISH to AnnData; spatial-graph builder); synthetic sanity check.
3–4	Spatial GNN (SAGE/GAT/GCN) + tuned XGBoost + MLP; shared leakage-safe trainer.
5	Cross-section, within-section, stratified experiments; multi-seed macro-F1/ARI.
6	Falsification ablations (graph removal, depth, neighbourhood size) on both tissues.
7	Biological interpretation (marker recovery, annotation-error flags); tumor-ST pilot.
8	Figures, tables, manuscript draft; reproducibility package.

Table 7: Schedule for the eight-week SRI cohort. The pilot covers the core pipeline, so the schedule focuses on validation, ablations, interpretation, and the tumor-ST extension.

12 Reproducibility and Ethics

Implementation uses open tools (PyTorch Geometric for the GNN; scanpy and anndata for spatial-omics I/O; XGBoost). All code, configs, random seeds, and a data manifest are released, and every reported number is multi-seed with standard deviations. No human subjects are involved; all inputs are public, de-identified post-mortem or model-organism tissue. Predictions are research-only and hypothesis-generating and do not constitute clinical guidance.

References

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